

Inequalities – 3

Integral inequalities: the working techniques

Homework

10 сентября 2026 года

Theoretical minimum. Chebyshev's integral inequality for similarly ordered functions; Jensen's inequality for integrals; normalisation; writing the difference of two sides as $\int(\cdot)^2$ or as a double integral with a fixed sign; integration by parts as a source of inequalities; introducing a parameter and optimising in it at the end.

1. (IMC 2005) Let $f: \mathbb{R} \rightarrow [0, \infty)$ be continuously differentiable. Prove that $\int_0^1 f^3 - f^2(0) \int_0^1 f \leq \max_{0 \leq x \leq 1} |f'(x)| \cdot \left(\int_0^1 f\right)^2$.
2. (IMC 2019) Let $f: (-1, 1) \rightarrow \mathbb{R}$ be twice differentiable with $2f'(x) + xf''(x) \geq 1$ on $(-1, 1)$. Prove that $\int_{-1}^1 xf(x) dx \geq \frac{1}{3}$.
3. (IMC 2004) Let $f, g: [a, b] \rightarrow [0, \infty)$ be continuous and non-decreasing with $\int_a^x \sqrt{f} \leq \int_a^x \sqrt{g}$ for every $x \in [a, b]$ and $\int_a^b \sqrt{f} = \int_a^b \sqrt{g}$. Prove that $\int_a^b \sqrt{1+f} \geq \int_a^b \sqrt{1+g}$.
4. (Putnam 2003) Let f be continuous on $[0, 1]$. Show that $\int_0^1 \int_0^1 |f(x) + f(y)| dx dy \geq \int_0^1 |f(x)| dx$.
5. (Putnam 1999) Prove that there is a constant C such that $|p(0)| \leq C \int_{-1}^1 |p(x)| dx$ for every polynomial p of degree 1999.
6. (RMO 2009) Find all $f \in C^1[0, 1]$ with $f(1) = -\frac{1}{6}$ and $\int_0^1 (f'(x))^2 dx \leq 2 \int_0^1 f(x) dx$.
7. (RMO 2021) Let $f: [0, 1] \rightarrow [0, 1]$ be continuous and bijective with $f(0) = 0$. Prove that $(\alpha + 2) \int_0^1 x^\alpha (f(x) + f^{-1}(x)) dx \leq 2$ for every $\alpha \geq 0$.

8. (RMO 2023) Let $f: [0, 1] \rightarrow \mathbb{R}$ be non-decreasing of class C^1 with $f(0) = 0$, and put $g(x) = f(x) + (x - 1)f'(x)$.
- Show that $\int_0^1 g = 0$.
 - Prove that $\int_0^1 g(\varphi(t)) dt \leq 0$ for every convex differentiable $\varphi: [0, 1] \rightarrow [0, 1]$ with $\varphi(0) = 0$ and $\varphi(1) = 1$.
9. (RMO 2024) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(x) + \sin f(x) \geq x$ for all x . Prove that $\int_0^\pi f(x) dx \geq \frac{\pi^2}{2} - 2$.
10. (Сибирская 2019) Let f be continuous and strictly increasing on $[a, b]$. Prove that $\frac{1}{c-a} \int_a^c f < \frac{1}{b-a} \int_a^b f$ for every $c \in (a, b)$, and give the one-line proof through the mean value of a monotone function.
11. (SEEMOUS 2015) Prove that $\int_0^1 \sqrt{1 + \cos^2 y} dy > \sqrt{x^2 + \sin^2 x}$ for every $x \in (0, 1)$, and explain the geometric meaning of both sides as lengths.